

ALY 6040 Data Mining Applications

Module 3 - Discussion

Shraddha Gupte



CPS - MPSA, Northeastern University

Prof. Arasu Narayan

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A. Clustering and Classification Models

Clustering and classification are two distinct types of machine learning models, though they can sometimes complement each other.

1. Difference between Clustering and Classification Models

- **Clustering Models:**

Clustering is a type of unsupervised learning where the goal is to group similar data points together without having predefined labels or categories. The objective of clustering is to find patterns or natural groupings in the data. The model assigns each data point to one of several clusters based on its characteristics.

Examples:

- **DBSCAN:** A density-based clustering method that groups data points based on the density of their surroundings, useful for finding irregularly shaped clusters.
- **Example Use Case:** Customer segmentation in marketing, where customers are grouped into clusters based on purchasing behavior or demographics without predefined labels.

- **Classification Models:**

Classification is a supervised learning task where the model learns to categorize data points into predefined classes or labels based on input features. The goal is to train the model to correctly predict the class or label for new, unseen data based on patterns learned during training.

Examples:

- **Logistic Regression:** Predicts binary outcomes (e.g., whether an email is spam or not).
- **Random Forest:** Uses multiple decision trees to classify data points into categories (e.g., whether a patient has a disease based on symptoms).
- **Example Use Case:** Image classification, where images are labeled as 'cat', 'dog', or 'other'.
- Clustering can be used to discover inherent structure in the data before classification.
- For instance, in semi-supervised learning, clustering may group similar data points together. Labels are then assigned to one or more clusters, allowing for classification based on these clusters.
- Clustering results can be used as additional features in a classification task. This can help in identifying patterns that are not directly visible in the raw data.

Example:

In fraud detection, clustering methods like DBSCAN might be used to identify abnormal spending behaviors (outliers). These behaviors can be treated as a feature when building a classification model to predict whether a transaction is fraudulent.

- After using classification, clustering can be used to analyze and better understand the subgroups within each class.

Example:

In medical diagnosis, after classifying patients as having a particular condition, clustering could reveal subgroups within this class based on factors like age, lifestyle, or severity of symptoms.

While clustering is not a direct method for classification, it can be used to enhance classification models by identifying structures, serving as features, or providing additional insights.

B. K-Means and Hierarchical Clustering

1. Difference Between K-Means and Hierarchical Clustering

- **K-Means Clustering:**

- K-Means is a partitioning-based clustering algorithm. It divides the dataset into a predefined number of clusters (K) by assigning data points to clusters based on the distance to the cluster's centroid.
- Initially, K centroids are randomly chosen. Each data point is assigned to the nearest centroid.
- Centroids are recalculated based on the mean of the points in each cluster, and the process is repeated until the centroids no longer change.
- It creates non-overlapping clusters, where each data point belongs to exactly one cluster.
- K-Means is computationally efficient, especially for large datasets, as it scales linearly with the number of data points.
- K-Means tends to work well when clusters are spherical and similarly sized because it relies on the Euclidean distance to determine clusters.

- **Example Use Case:**

Image compression: By reducing the number of colors in an image (by grouping similar colors into K clusters), K-Means can significantly compress an image while maintaining visual quality.

- **Use K-Means When:**

- There is a large dataset, as it is computationally efficient and can handle large volumes of data well.
- We already know the number of clusters (K) or have a good estimate of how many clusters there should be.
- The clusters are expected to be approximately spherical and equally sized.
- Speed and simplicity are more important than precision, and we want a fast algorithm to partition data.

- **Hierarchical Clustering:**

- Hierarchical clustering is a hierarchy-based or tree-based method. It doesn't require the number of clusters to be specified upfront and forms a dendrogram (a tree-like structure) of nested clusters.
- It starts by treating each data point as its own cluster. Then, it merges the closest clusters step by step, creating a hierarchy of clusters until all data points are in one cluster.
- The reverse is also possible, where all points start in one large cluster and are successively split.
- It produces a nested hierarchy of clusters, allowing for subclusters within larger clusters. This creates a tree structure that shows the relationships between clusters at different levels.
- Hierarchical clustering can be computationally expensive, especially for large datasets, as it requires calculating distances between every pair of data points. This can be slow for datasets with thousands of points.
- It can handle clusters of arbitrary shapes and sizes because it doesn't rely on specific geometric assumptions like K-Means does.

- **Example Use Case:**

Customer segmentation: In hierarchical clustering, customer segments can be nested, revealing subgroups within larger groups. This is useful when we want a deeper understanding of the relationships among customer segments.

- **Use Hierarchical Clustering When:**

- We need to visualize the data structure or relationships between clusters at different levels using a dendrogram.
- The number of clusters is unknown, or we want flexibility to determine the number of clusters by cutting the dendrogram at different levels.
- We are dealing with small to moderately sized datasets, as it is computationally more expensive for large datasets.
- Clusters are not spherical or we expect nested clusters or clusters of different shapes and sizes.

1. Summary of Key Differences

Feature	K-Means Clustering	Hierarchical Clustering
Algorithm Type	Partitioning-based	Hierarchy-based (agglomerative or divisive)
Clusters Shape	Assumes spherical clusters	Can handle arbitrary shapes
Number of Clusters	Must be specified before running the algorithm	Automatically determined, can cut dendrogram
Computational Cost	Efficient for large datasets	More expensive, scales poorly for large datasets
Output	Non-overlapping clusters	Nested clusters (dendrogram)
Use Case	Large datasets with clear K	Smaller datasets, nested or complex clusters

C. References

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